

Simplex Method

Maximize $Z = x_1 + 4x_2$

Constraints

$$2x_1 + x_2 \leq 3$$

$$3x_1 + 5x_2 \leq 9$$

$$x_1 + 3x_2 \leq 6$$

$\left. \begin{array}{l} \text{--- } s_1 \\ \text{--- } s_2 \\ \text{--- } s_3 \end{array} \right\}$

$$x_1, x_2 \geq 0$$

Introduce the slack variables
related to the number of
unknowns.

Unknowns are ~~x_1, x_2~~

$$\text{Max } Z = x_1 + 4x_2 - 0s_1 - 0s_2 - 0s_3$$

$$Z - x_1 - 4x_2 + 0s_1 + 0s_2 + 0s_3 = 0$$

$$2x_1 + x_2 + s_1 + 0s_2 + 0s_3 = 3$$

$$3x_1 + 5x_2 + 0s_1 + s_2 + 0s_3 = 9$$

$$x_1 + 3x_2 + 0s_1 + 0s_2 + s_3 = 5$$

$$x_1, x_2, s_1, s_2 \geq 0$$

Basic Variables		Coefficients					Value	Ratio
		x_1	x_2	s_1	s_2	s_3		
$R_1 \rightarrow Z$		-1	-4	0	0	0	0	$9/3 = 3$
R_2	s_1	2	1	1	0	0	3	$3/1 = 3$
R_3	s_2	3	5	0	1	0	9	$9/5 = 1.8$
R_4	s_3	1	3	0	0	1	5	$5/3 = 1.67$

Put the min value.

Key Row

$$\frac{1}{3} \quad \frac{3}{3} \quad \frac{0}{3} \quad \frac{0}{3} \quad \frac{1}{3} \quad \frac{5}{3}$$

$$\frac{4}{3}$$

$$1$$

$$0$$

$$0$$

$$\frac{1}{3}$$

$$\frac{5}{3}$$

To identify the kth, we
use the Z-Value
Rows with the
Smallest Value. $\rightarrow$

from s_1, s_2 & s_3 , Pick the
middle value, w/c is 3.

Using The Key Row, I want
to make all the elements
in the Key Column Zero.
Use Row Reduction Using
selcon.

Basic Var → Z	x_1	x_2	s_1	s_2	s_3	RHS	Ratio
	$\frac{1}{3}$	0	0	0	$\frac{4}{3}$	$\frac{27}{3}$	
s_1	$\frac{5}{3}$	0	1	0	$-\frac{1}{3}$	$\frac{4}{3}$	
s_2	$\frac{1}{3}$	0	0	1	$-\frac{5}{3}$	$\frac{7}{3}$	
s_3	$\frac{1}{3}$	1	0	0	$\frac{1}{3}$	$\frac{5}{3}$	

Key Column.

$$R_2 - 5R_4, \quad = 3 - 5\left(\frac{1}{3}\right)$$

$\uparrow$
 R_3

$$= \frac{2-5}{3} = -\frac{1}{3}$$

$$5 - 5(1) = 0$$

$$= 9 - 5\left(\frac{5}{3}\right)$$

$$= \frac{27-25}{3} = \frac{2}{3}$$

$$R_2' = R_2 - R_4$$

$$2 - \frac{1}{3} = \frac{6-1}{3} = \frac{5}{3}$$

$$R_1' = R_1 + 4R_4$$

$$-4 + 4(1) = \underline{\underline{0}}$$

$$R_1' = -1 + 4\left(\frac{1}{3}\right) = -1 + \frac{4}{3} = \frac{1}{3}$$

To get the Maximum Value ↓

$$Z_{\max} = x_1 + 4x_2 = \frac{20}{3}$$

from RHS, $s_1, s_2, \& s_3$,
when

$$s_3 = x_2 = \frac{5}{3}, x_1 = 0$$

$$Z_{\max} = \frac{20}{3} \text{ (satisfied)}$$